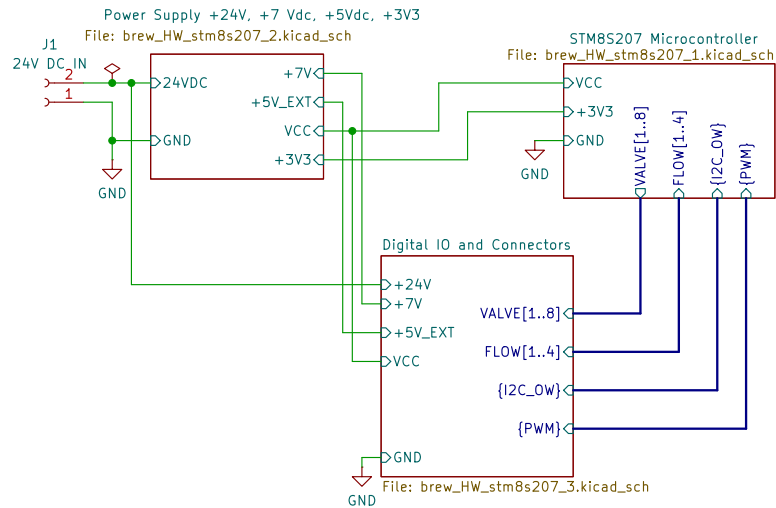


- Features:
- 6-8 Temperature-sensors (2 x I2C, 4-6 0W sensors)
 - 4 x Flow-sensor
 - 2 x 230V gasburner enable + 2 x PWM modulating control
 - 2 x 3 phase electric SSR for HLT and BK
 - 2 x SSR for Pumps
 - 8 x Solenoid Ball Valve control
 - Ethernet RJ45 by Wiz550io
 - Frontpanel LED control by MAX7219



- 2.0: First version. PCB made by IteadStudio.
2.1: Bug-fix PWM generation, R48 and R49 added.
3.0: Major update: triacs, pcf8574, max1238, con3 (valves) removed, added: flow sensor interface + valves interface.
3.01: IteadStudio PCB-2 made
3.02: JP1 for Int./Ext. 24AC switching added
MCP23008 with 2 ULN2803 parallel for 1A switching
3.03: 2 x DS2482 / One-Wire Interface added
3.04: PTC (Resettable Fuses) added to Valve Interface
3.05 WIZ550io Interface added
3.10 Pressure Sensor Interface / -10V removed
Sensor Connectors now DIN6-connectors.
FLOW2 from D2 to A2, FLOW3 (A1) added.
+10V now with DC-DC converter.
3.20 All Resistors now 1206 SMD. Arduino Nano instead of UNO.
LM35 from A0 to A6, FLOW4 added to ADC0.
Two One-Wire temp. sensors added (IC11, IC12, CON16, CON17).
3.21 Temp. sensors now with RJ12 connectors.
3.30 IC MCP23017 added + Boil PWM interface added.
Heater_LED now via triac pcb. SPLSS from D10 to D7 (PD7).
3.40 LD1117 footprint added + TD62783 instead of ULN2803.
PUMP2_230V added.
3.50 Solenoid intf now with relays, SMD footprints when possible.
3.60 VCC-extern now separate from VCC.
4.00 Arduino replaced by stm8s207R8
4.01 BK_SSR, HLT_SSR and FLOW4 to other pins
4.10 Level-shifter MISO + 2x3 ELE SSR added
4.11 CH340G + USB added
4.12 Bug-fix USB connector + DC PWR changed to 24V and 7V + 3V3 separate from 5V
4.20 KiCad 9 Conversion
4.21 C50 value 100n->820n
U6 address should be 0x36
CON7 pin 1 had no Vcc connected
TX2/RX2 should be RX3/TX3, also on CON19
R44/R51 removed, R25: 10K->18K, C21 added.
LM393 / inverted PWM -> normal PWM.
4.30 KiCad 10 Conversion
Wiz550IO orientation changed
W5500_Lite added

- H4 MountingHole
- H3 MountingHole
- H2 MountingHole
- H1 MountingHole

HERMS Brew Hardware Top-Level Design

Sheet: /
File: brew_HW_stm8s207.kicad_sch

Title: Brew Hardware STM8S207R8

Size: A4 Date: 2026-09-27

KiCad E.D.A. 10.0.5

Rev: R4.30

Id: 1/4

